

MAUKE, Cook Islands

WMO: 918400

Lat: 20.137S

Lon: 157.344W

Elev: 30

StdP: 14.68

Time Zone: -10.00 (W10)

Period: 82-96

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	65.4	66.9	54.2	62.4	70.3	55.6	65.8	70.6	23.6	72.7	21.2	72.4	2.6	130	0.309

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	8.1	87.3	79.4	86.5	79.0	85.7	78.6	80.9	85.2	80.2	84.6	79.7	84.0	7.4	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.6	153.6	83.4	79.0	150.7	83.1	78.5	148.3	82.8	44.6	85.7	44.0	85.0	43.3	83.9	83.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.4	18.2	15.5		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	77.4	80.2	81.1	81.3	79.5	77.4	75.4	74.3	73.9	74.5	75.7	77.5	78.5
	DBStd	3.44	1.73	1.72	1.69	2.21	2.22	2.30	2.96	2.75	2.62	2.45	2.09	2.10
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	10013	936	872	970	886	850	762	754	741	734	798	826	884
	CDD65	4538	471	452	505	436	385	312	289	276	284	333	376	419
	CDH74	32015	4346	4531	5002	3738	2452	1361	1105	962	1096	1658	2510	3254
	CDH80	5164	823	1116	1232	703	225	73	60	41	63	124	272	432
(14) Wind	WSAvg	7.2	7.0	6.4	6.0	6.8	7.5	7.1	7.2	7.8	7.0	7.8	7.8	7.8
(15) Precipitation	PrecAvg	64.8	8.9	7.9	7.2	7.0	6.1	2.9	3.4	3.2	2.8	3.9	4.4	7.1
	PrecMax	85.9	30.9	23.1	19.9	14.0	19.5	8.5	12.2	11.5	9.3	9.2	10.7	17.7
	PrecMin	26.9	0.4	1.1	1.0	1.3	0.9	0.9	0.6	0.8	0.8	0.3	0.7	1.2
	PrecStd	13.5	6.3	5.0	4.4	3.0	3.9	1.9	2.5	2.2	2.0	2.2	2.6	3.8
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	87.2	88.4	88.6	87.3	84.9	83.3	83.3	82.8	83.1	84.1	85.7	85.5
		MCWB	79.5	79.7	79.6	79.7	78.0	75.5	76.3	75.8	75.2	76.2	78.0	77.7
	2%	DB	86.2	87.3	87.5	86.2	83.6	81.9	81.7	81.2	81.7	82.7	84.0	84.7
		MCWB	79.1	79.4	79.4	78.9	77.2	75.1	75.2	74.7	74.8	75.4	76.8	77.2
	5%	DB	85.3	86.3	86.4	85.1	82.6	80.7	80.5	80.0	80.4	81.5	82.8	83.7
		MCWB	78.6	79.0	79.1	78.1	76.5	74.0	74.2	74.0	73.9	74.7	75.9	76.6
	10%	DB	84.1	85.4	85.4	83.8	81.4	79.4	79.1	78.6	79.0	80.3	81.7	82.6
		MCWB	78.0	78.5	78.7	77.3	75.8	73.4	73.4	73.3	73.0	73.9	75.1	75.9
	0.4%	WB	81.0	81.7	81.5	81.2	79.2	77.6	78.4	77.3	77.8	77.8	79.5	80.8
		MCDB	84.8	86.2	86.3	85.4	82.4	81.2	81.7	80.8	80.9	82.0	83.5	83.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	80.2	80.7	80.7	80.3	78.3	76.9	76.8	76.3	76.7	76.8	78.3	79.3
		MCDB	84.2	85.7	85.4	84.5	81.7	80.4	79.7	79.7	80.1	80.9	82.3	82.3
	5%	WB	79.7	79.9	80.0	79.5	77.5	76.1	75.7	75.5	75.6	76.0	77.6	78.3
		MCDB	83.6	84.7	84.7	83.4	81.0	79.6	79.2	78.9	79.1	80.0	81.7	81.7
	10%	WB	79.0	79.1	79.4	78.6	76.8	74.9	74.7	74.5	74.4	75.1	76.7	77.6
		MCDB	82.8	83.7	84.0	82.4	80.4	78.6	78.4	77.9	78.1	79.1	80.7	81.0
(35) Mean Daily Temperature Range	5% DB	MDBR	7.3	8.1	8.3	7.8	7.3	7.9	8.1	7.9	8.2	7.7	7.7	7.5
		MCDBR	8.7	9.4	9.6	8.9	8.2	9.0	9.3	9.5	9.5	8.6	8.6	8.5
		MCWBR	4.2	4.3	4.2	4.1	4.5	4.8	5.3	5.2	4.8	4.6	4.5	4.5
	5% WB	MCDBR	7.9	8.2	8.9	7.5	7.1	7.8	8.2	8.2	8.5	8.0	8.0	7.2
		MCWBR	4.4	4.4	4.5	4.3	4.5	5.2	6.0	5.5	5.2	5.0	5.1	4.3
(40) Clear-Sky Solar Irradiance	taub	0.379	0.384	0.376	0.380	0.374	0.365	0.362	0.362	0.362	0.382	0.376	0.377	0.382
	taud	2.637	2.618	2.654	2.636	2.634	2.636	2.632	2.632	2.626	2.564	2.587	2.608	2.613
	Ebn at Noon	306	301	296	282	272	269	273	284	290	300	305	305	305
	Edh at Noon	32	32	30	29	27	25	26	28	32	33	33	33	33
(44) All-Sky Solar Radiation	RadAvg	1996	1946	1781	1515	1262	1202	1262	1488	1772	1927	2031	1995	
(45)	RadStd	161	147	152	100	103	65	74	96	90	161	149	156	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page